NETWORK SERVICES

EXERCISES

Exercise 1:

A company has been assigned the following network:

192.168.10.0/24

The network administrator wants to divide this network into 4 equal-sized subnets to serve four departments.

Questions

1. How many bits must be borrowed from the host portion?
2. What is the new subnet mask in:
    - CIDR notation?
    - Dotted-decimal notation?
3. How many subnets will be created?
4. How many usable host addresses are available in each subnet?
5. Complete the following table:

Subnet	Network address	First usable IP	Last usable IP	Broadcast
1	?	?	?	?
2	?	?	?	?
3	?	?	?	?
4	?	?	?	?

6. Which subnet contains the IP address 192.168.10.150?
7. Can the IP address 192.168.10.128 be assigned to a PC? Justify your answer.

Exercise 2:

A university has received the network address:

192.168.20.0/24

The network administrator must divide this network into four subnets according to the following requirements:

Department	Required number of hosts
Administration	100
Computer Science	50
Research Laboratory	25
Guest Network	10

Questions

1. Explain why VLSM is more appropriate than using the same subnet mask for all departments.
2. Sort the departments from the largest to the smallest number of required hosts.
3. For each department, determine:
    - The minimum number of host bits required.
    - The appropriate CIDR prefix.
    - The subnet mask.
4. Allocate an appropriate subnet to each department, starting from 192.168.20.0.
5. Complete the following table:

Department	Network address	Prefix	First usable IP	Last usable IP	Broadcast
Administration	?	?	?	?	?
Computer Science	?	?	?	?	?
Research Lab	?	?	?	?	?
Guest Network	?	?	?	?	?

6. Calculate the remaining unused IP address space.
7. Explain one advantage of VLSM in enterprise network design.

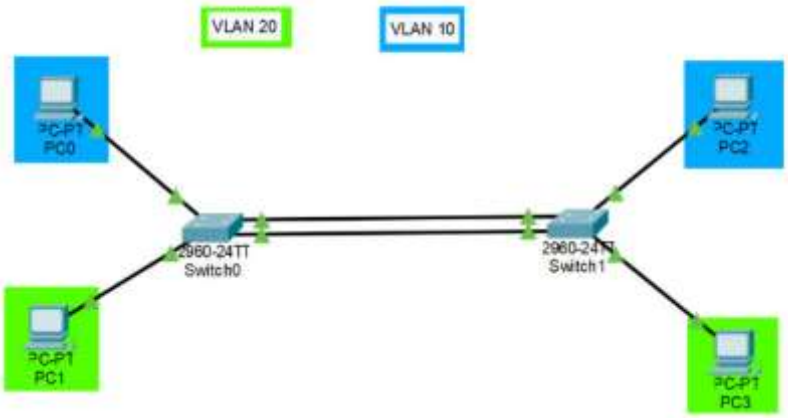
Important: Allocate the largest subnet first and ensure that subnet ranges do not overlap.

Exercise 3:

A company has two departments:

- Administration
- Engineering

The company wants to logically separate these departments using VLANs.



The network consists of:

- Two Cisco 2960 switches (S1 and S2).
- Four PCs (PC1, PC2, PC3 and PC4).

- One trunk connection between the two switches.

IP addressing plan

Device	Switch	VLAN	IP address
PC1	S1	10 - ADMIN	192.168.10.10/24
PC2	S1	20 - ENGINEERING	192.168.20.10/24
PC3	S2	10 - ADMIN	192.168.10.20/24
PC4	S2	20 - ENGINEERING	192.168.20.20/24

Questions

1. Create VLAN 10 named ADMIN and VLAN 20 named ENGINEERING on both switches.
2. Assign the following access ports:
    - S1: Fa0/1 → PC1, Fa0/2 → PC2.
    - S2: Fa0/1 → PC3, Fa0/2 → PC4.
3. Configure the link between S1 and S2 as a trunk using IEEE 802.1Q.
4. Allow only VLANs 10 and 20 on the trunk.
5. Verify the configuration using:
    - show vlan brief
    - show interfaces trunk
6. Test connectivity using the ping command:

    a. PC1 → PC3

    b. PC2 → PC4

    c. PC1 → PC2
7. Explain why PC1 can communicate with PC3 but cannot communicate directly with PC2.